PHARMACY MANAGEMENT SYSTEM

A Centralized Web-Based ERP Platform for Retail Pharmacy Management

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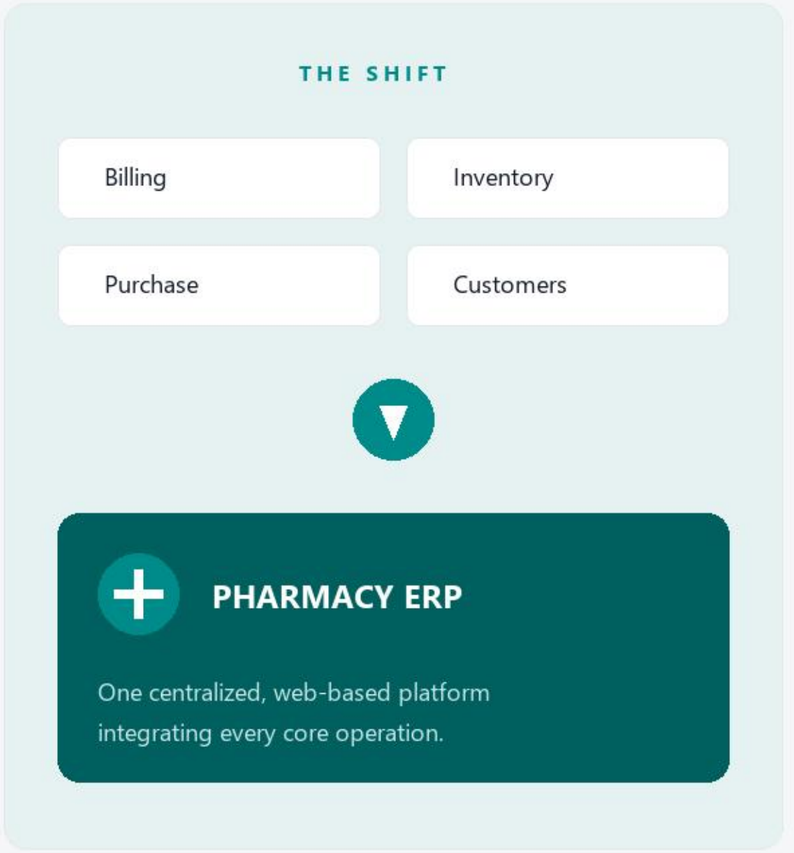
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INTRODUCTION

From Fragmented Operations to One Platform

- Pharmaceutical retail involves interconnected operations — billing, inventory, purchasing, customer management, and financial monitoring.
- Traditional pharmacy operations often rely on manual records or fragmented, disconnected software.
- This fragmentation makes stock monitoring, expiry tracking, billing, and business analysis difficult.
- PHARMACY ERP provides a centralized, web-based ERP platform that integrates these operations.
- The system is purpose-built around the real operational needs of a pharmacy.



PROBLEM STATEMENT

Problems in Traditional Pharmacy Management

01

Inventory Expiry

Medicines may sit in stock until expiry; lack of batch-level monitoring causes financial losses.

02

Billing Inefficiency

Manual medicine search and billing is slow, and tax calculation is prone to human error.

03

Stock Management

Individual batches are hard to track, and low-stock medicines often go unnoticed.

04

Customer Management

Repeated manual customer entry and poor access to previous purchase history.

05

Financial Monitoring

Limited real-time visibility into revenue, profit, and daily cash/UPI sales.

06

Access Control

Different employees need different levels of system access — rarely enforced.

OBJECTIVES

Objectives of Pharmacy ERP

- Develop a centralized, web-based Pharmacy ERP
- Implement an interactive Point of Sale (POS) system
- Provide batch-level inventory management
- Implement FEFO (First-Expire-First-Out) handling
- Automate GST-aware billing calculations
- Maintain customer information & purchase history
- Provide real-time business intelligence dashboards
- Identify low-stock and expiry-related risks
- Implement role-based access control (RBAC)
- Deploy the application on a cloud-hosted environment

EXISTING SYSTEM

The Conventional Approach

- Medicine Purchase
- Manual / Separate Records
- Stock Management
- Billing
- Customer Records
- Financial Calculation
- Manual Reporting

MAJOR DRAWBACKS

- Fragmented information
- Manual data entry
- Difficult batch-level tracking
- Expiry monitoring challenges
- Slower billing workflow
- Limited real-time analytics
- Manual procurement monitoring
- Limited role-based access



MAIN GAP

Operational data exists — but it is not sufficiently integrated.

PROPOSED SYSTEM

PHARMACY ERP — Core Modules

Integrates every major pharmacy operation into one centralized platform.

POS & Billing

- Medicine search
- Barcode input
- Cart management
- Billing

Inventory

- Batch tracking
- Expiry management
- FEFO
- Low-stock monitoring

CRM

- Customer directory
- Purchase history
- AJAX-based lookup
- Lifetime spend metrics

Financial & BI

- GST calculation
- Revenue tracking
- Profit calculation
- Sales analytics

Security

- Role-based access
- Financial protection
- Session auto-close
- Admin vault route

SYSTEM MODELING

Data Flow Diagram — Level 0 (Context Diagram)



Level 0 Scope: Encapsulates human interaction roles, wholesale supply-chain, and POS transaction boundary.

SYSTEM MODELING

Data Flow Diagram — Level 1 (Decomposition Diagram)

1.0 **Authentication & RBAC Control**
Validates login against User/Group models; routes to Admin, Manager, or Cashier UI.
[Auth_User]

2.0 **POS & Reverse GST Billing Engine**
Accepts barcode/search input; calculates GST dynamically; creates Sale record.
[D1, D2 -> D3]

3.0 **FEFO Batch Inventory Dispatch**
Sorts stock by expiry_date; deducts sold units; triggers Signals on deletion.
[D2 Stock]

4.0 **Reverse Logistics (Returns Desk)**
Scans receipt barcode; validates bill items; restocks physical units; logs audit.
[D3 -> D2, D4]

5.0 **Smart CRM & Customer Directory**
Executes AJAX telephone lookup; links customer to doctor and spend total.
[D5 Customer]

6.0 **Procurement & Reorder Engine**
Evaluates batch stocks against minimum_stock_level; exports CSV reorder PO.
[D1, D2 -> PO]

7.0 **Financial BI & Analytics Engine**
Aggregates daily sales, profit margins, GST totals, and payment modes.
[D3 Ledger]

DATABASE DATA STORES (RELATIONAL REPOSITORY)

D1 **Medicine Catalog Table**
Attributes: barcode, name, category, pack_size, gst%, min_stock.

D2 **Stock Batches Table**
Attributes: medicine_id, batch_no, expiry_date, quantity, purchase_price, mrp.

D3 **Sale Ledger Table**
Attributes: bill_no, stock_id, quantity, total_price, customer_id, payment_method, date.

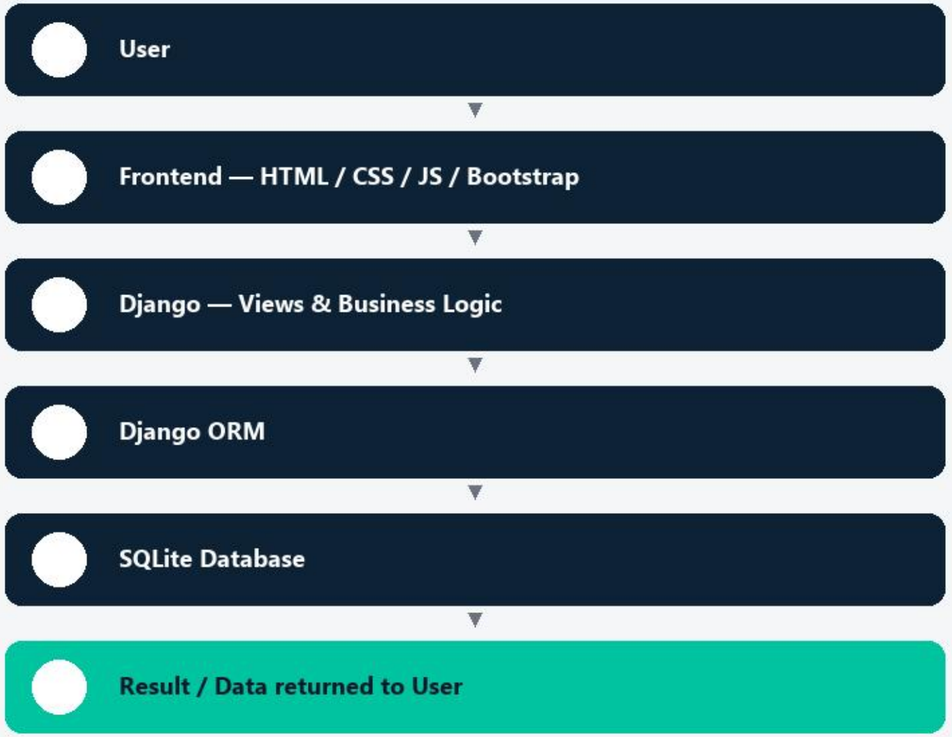
D4 **ReturnSale Audit Table**
Attributes: sale_id, stock_id, quantity_returned, refund_amount, reason, restocked.

D5 **Customer CRM Table**
Attributes: name, phone (unique), last_doctor, cumulative purchase history.

D6 **Supplier Directory Table**
Attributes: name, contact_person, phone, email, address.

METHODOLOGY

PHARMACY ERP — Working Model

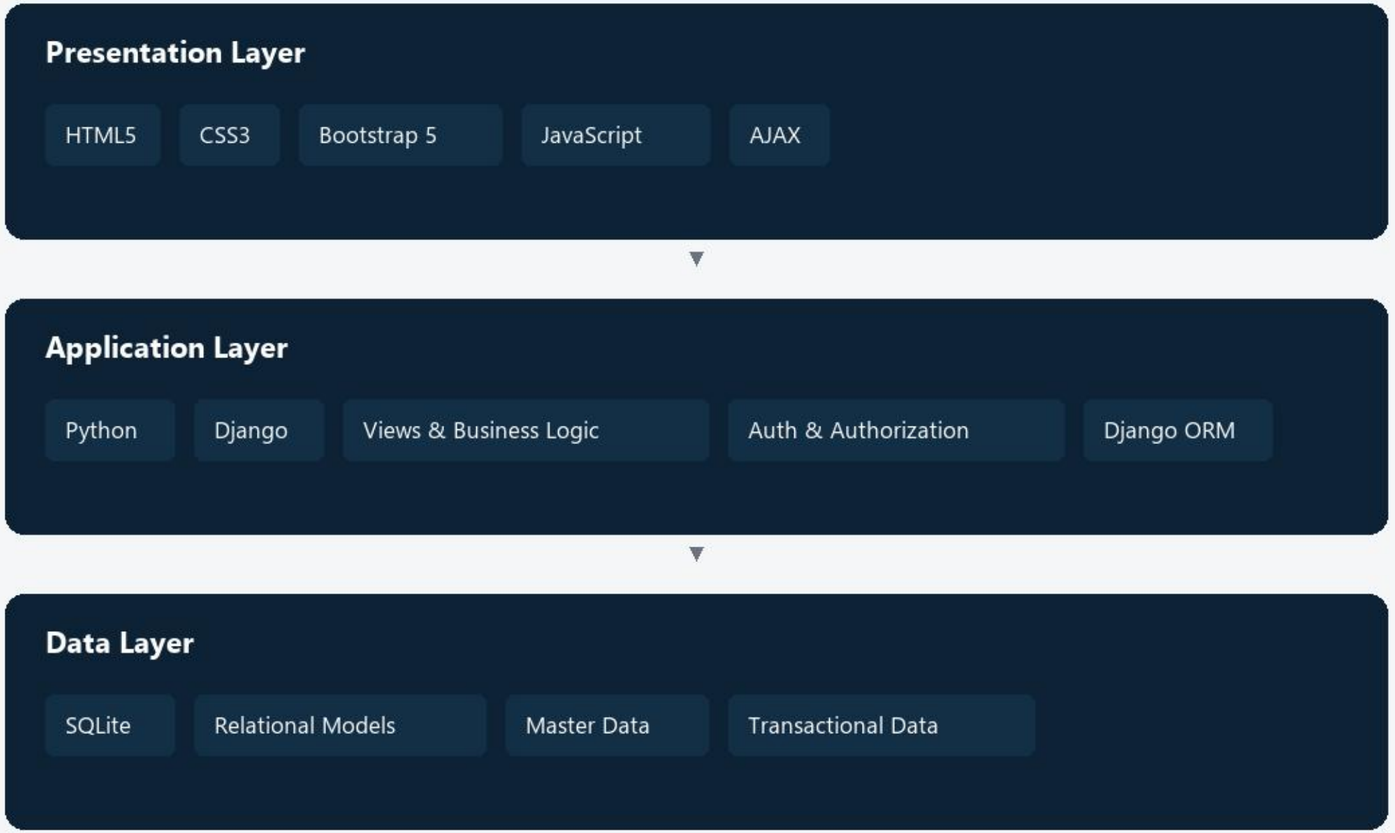


EXAMPLE — BILLING WORKFLOW

1. Select / scan medicine
2. Retrieve batch information
3. Apply FEFO logic
4. Calculate GST
5. Create sale record
6. Update inventory
7. Generate bill

SYSTEM ARCHITECTURE

Three-Tier Modular Architecture



SUPPORTING TECHNOLOGIES

- Select2
- JsBarcode
- Chart.js
- python-dotenv

DEPLOYMENT

- PythonAnywhere
- Django WSGI

TOOLS & TECHNOLOGIES

Technology Stack

Layer	Technologies
Programming Language	Python 3.14
Backend	Django 6.0.4
Database	SQLite
ORM	Django ORM
Frontend	HTML5, CSS3
UI Framework	Bootstrap 5
Client-side Scripting	Vanilla JavaScript (ES6)

Layer	Technologies
Asynchronous Requests	AJAX
Search	Select2
Barcode	JsBarcode
Visualization	Chart.js
Configuration	python-dotenv
Deployment	PythonAnywhere
Server Interface	Django WSGI

KEY TECHNICAL FEATURES

Where the Technical Depth Shows

FEFO Algorithm

Prioritizes earliest-expiring medicine batches automatically.

AJAX Operations

Enables asynchronous customer lookup and master-data creation.

Dynamic GST

Performs GST-aware billing calculations in real time.

Django Signals

Synchronizes inventory automatically when transactions change.

RBAC

Separates access between Owner, Manager, and Cashier roles.

Business Intelligence

Converts sales data into revenue, profit and inventory insights.

APPLICATIONS

Applications of Pharmacy ERP

PRIMARY APPLICATION

Current project scope

- Retail pharmacies
- Medical stores
- Independent pharmacy businesses

POTENTIAL EXTENDED APPLICATIONS

Future growth directions

- Multi-branch pharmacy management
- Pharmacy chain operations
- Centralized inventory monitoring
- Pharmacy-owner business analytics
- Mobile-based pharmacy management

PROJECT TIMELINE

Development Progression

Actual build sequence of the already-developed project (replace with real dates if available).

1. **Requirement Analysis**
2. **System Design & Database Modeling**
3. **Backend Development (Django)**
4. **Frontend & POS Development**
5. **Inventory & FEFO Implementation**
6. **CRM & AJAX Integration**
7. **BI Dashboard & Financial Modules**
8. **Security & RBAC**
9. **Testing & Bug Fixing**
10. **Cloud Deployment**
11. **Documentation & Presentation**

FUTURE SCOPE

Phase 2 Roadmap

AI & Machine Learning

- Prescription OCR
- Medicine demand forecasting
- Seasonal shortage prediction

Mobile & API

- Django REST Framework
- Mobile app for pharmacy owners
- Bluetooth receipt printer sync

Communication Automation

- WhatsApp Business integration
- Digital receipt delivery
- Automated supplier alerts

Cloud & DevOps

- PostgreSQL migration
- Docker containerization
- CI/CD pipeline
- Scalable cloud deployment

CONCLUSION

Pharmacy ERP, in Summary

- Provides an integrated ERP solution for end-to-end pharmacy operations.
- Combines billing, inventory, CRM, financial processing, analytics and security.
- Batch-level inventory management and FEFO improve stock handling.
- AJAX-based workflows improve operational efficiency.
- BI dashboards provide better business visibility.
- RBAC provides controlled access to system functionality.
- Cloud deployment makes the application accessible beyond local development.

PHARMACY ERP transforms fragmented pharmacy operations into a centralized digital management workflow.